BTS-III(S)-04.18-0791 Reg. No. | | | | | | | | |

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B.Tech. Degree III Semester Supplementary Examination April 2018

CS/IT 15-1303 DISCRETE COMPUTATIONAL STRUCTURES

(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer ***ALL*** questions)

$(10 \times 2 = 20)$

I. (a) Verify that given compound proposition is tautology or not

$$((p \to q) \wedge (p \to r)) \to (p \to r)$$

(b) If R is an equivalence relation on A = {1, 2, 3, 4, 5}. Prove that R^{-1} is also an equivalence relation on A.

(c) State Pigeon-Hole principle. Show that if 9 colours are used to paint 100 houses, at least 12 houses will be of same colour.

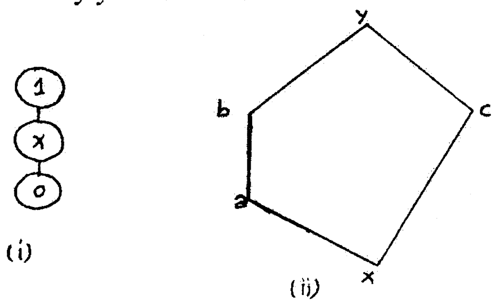
(d) Write an algorithm to find the maximum value in a finite sequence.

(e) Draw a 3-regular graph with ten vertices and a 4-regular graph with nine vertices.

(f) Define a connected graph. Draw the complete graph K_4 and examine whether it is a connected graph.

(g) Show that set of all positive rational numbers form an abelian group under the operation *defined by a*b = ab/2.

(h) Determine whether the posets shown in the figure are distributive lattices or not. Justify your answer.



(i) Differentiate between best case and worst case time of an algorithm.

(j) Represent the expression as a binary tree and perform post order traversal on the binary tree. A*B – (C + D) – E/F.

PART B

$(4 \times 10 = 40)$

II. (a) Using the principle of mathematical induction. Show that $1 + 3 + 5 + \ldots\ldots\ldots + 2n\text{-}1 = n^2$. (6)

(b) Let X = {1, 2, ... 7} and R = {<X, Y>/X-Y is divisible by 3}. Show that R is an equivalence relation. (4)

OR

III. (a) Let f, g and h be the functions defined on a set of positive integers defined by the equations $f(x) = x^2$, $g(x) = x - 1$ and $h(x) = x/2$. Find hogof, gofoh and fogoh. (5)

(b) Which of the following arguments are not valid? Justify your answer. (5)

(i) "If I get the job and works hard, then I will be promoted. If I get promotion, then I will be happy. I will not be happy, therefore, either I will not get the job or I will not work hard".

(ii) "Either Ram is not guilty or Sunil is telling the truth. Sunil is not telling the truth, therefore, Ram is not guilty".

(iii) If n is a real number such that n > 1, then $n^2 > 1$. Suppose that $n^2 > 1$, then n > 1.

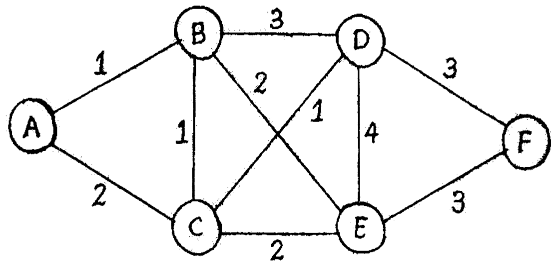
(P.T.O.)

IV. Solve recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$ with initial condition $a_0 = 1$ and $a_1 = 4$. (10)

OR

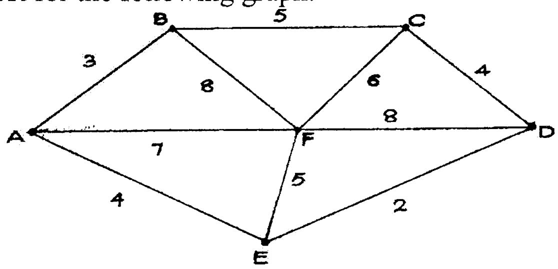
V. Solve recurrence relation $a_n = 4a_{n-1} - 3a_{n-2}$ with initial conduction $a_1 = 0$ and $a_2 = 12$. (10)

VI. Apply Dijkstra's algorithm for the following graph to find the shortest path. (10)



OR

VII. What is minimum spanning tree? How is the minimum cost spanning tree calculating using Kruskal's algorithm? Find the MSP and minimum cost for the following graph. (10)



VIII. Let D be the set of factors of a positive integer 'm' and $\leq$ be the relation divides. Draw the Hasse diagram for (a) m = 50 and (b) m = 100. (10)

(i) Determine the GLB and LUB of B where B = {10, 20}

(ii) Determine the GLB and LUB of B where B = {5, 10, 20, 25}

OR

IX. (a) Let a = {a, b}. Which of the following tables defines a semigroup on A? Which defines a monoid on A? (7)

(i)

*	a	b
a	a	b
b	b	a

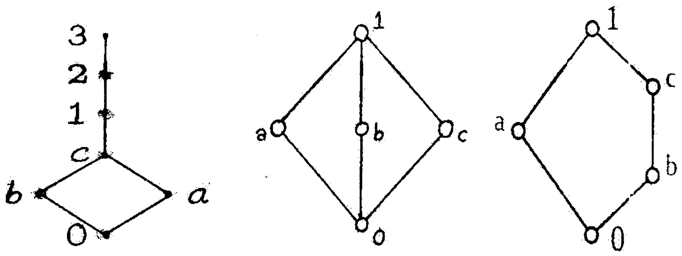
(ii)

*	a	b
a	a	a
b	b	a

(iii)

*	a	b
a	b	b
b	a	b

(b) Determine whether the posets shown in the figure are bounded lattices or not. Justify your answer. (3)



(i) (ii) (iii)

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